SALESFORCE ADMIN COURSE • EPISODE 34

Profiles, Permission Sets & Permission Set Groups

The Complete Guide to Salesforce Identity & Access Architecture: Baselines, Additive Privileges, Persona-Based Packaging, and Muting Logic.

- 👤 Profile = Single Baseline Access (1:1 Rule)
- ➕ Permission Sets = Additive Tasks & Privileges
- 📦 Permission Set Groups = Persona Bundles
- 🔇 Muting Permission Sets = Selective Removal
- 🛡️ Minimum Access Baseline Strategy
- 💉 15-Step End-to-End Hands-On Demonstration

Created by Sachin • Salesforce Certified Administrator Series • Duration: ~28–32 Mins

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1. Introduction & The Core Access Control Question

In previous episodes, we established our foundational perimeter security: creating Users, setting Password Policies, locking down Login Hours, enforcing Login IP Ranges, and requiring Multi-Factor Authentication (MFA).

Those layers answered the question: **"Can this user securely log in to Salesforce?"**

Now, we transition into internal authorization and privileges: **"What can this user actually see and do once inside Salesforce?"**

👤 Scenario: Standard Sales Rep (Rahul)

Needs everyday CRM capabilities:

- View Accounts & Contacts
- Create and update Leads
- Edit pipeline Opportunities
- **No Delete privileges**
- **No Report Export privileges**

👔 Scenario: Sales Manager / Lead (Priya)

Needs elevated managerial control:

- Standard pipeline management
- Permission to delete duplicate Opportunities
- Permission to Export corporate Reports to Excel/CSV
- Bulk data maintenance access

How do administrators support differing user needs without creating a logistical maintenance nightmare? Salesforce gives us three tightly integrated tools:

1. PROFILE

Baseline Access & Default Settings (Mandatory: Exactly 1 per User)

⬇️ **extends with**

2. PERMISSION SET

Atomic, Task-Based Additive Permissions (Optional: 0, 1, or Many per User)

⬇️ **packaged into**

3. PERMISSION SET GROUP

Bundled Sets Assigned by Persona/Job Function (Optional: 0, 1, or Many per User)

2. What is a Profile? (The Baseline Access & The 1:1 Rule)

A **Profile** defines a user's fundamental baseline access and default configuration settings within Salesforce. When any user account is provisioned, selecting a Profile is mandatory.

> **The Absolute Cardinal Rule of Profiles:**
> Every active user in Salesforce has **EXACTLY ONE Profile** assigned at any given moment. You cannot assign Rahul both the *Sales User Profile* and the *Sales Manager Profile* simultaneously.



> **Modern Salesforce IAM Best Practice:**
> Salesforce officially recommends keeping Profiles as *minimal as possible* (using the standard `Minimum Access - Salesforce` profile as a baseline template) and using **Permission Sets** and **Permission Set Groups** to distribute all business permissions and object/field access!

3. What Can a Profile Control? (9 Key Capabilities)

A Profile acts as the central control plane for a broad range of settings, permissions, and interface layouts:

1. System Permissions

Global capabilities such as *View Setup and Configuration*, *Export Reports*, *Manage Users*, and API Access.

2. Object Permissions

CRUD + View All / Modify All settings (e.g. Can users Create, Read, Edit, or Delete Opportunities?).

3. Field Permissions (FLS)

Controls field-level security for every standard and custom field (*Read Access* and *Edit Access*).

4. App Access & Defaults

Controls which Lightning/Classic apps appear in the App Launcher and which app opens by default on login.

5. Tab Settings

Determines tab visibility on navigation bars: *Default On*, *Default Off*, or *Tab Hidden*.

6. Record Types

Controls which Record Types are selectable when creating records and assigns the user's default Record Type.

7. Page Layouts

Maps which Page Layout is rendered for each Object and Record Type combination.

8. Login Hours

Specifies authorized login days and hours (e.g. Mon–Fri, 9:00 AM – 6:00 PM). Denies access outside windows.

9. Login IP Ranges

Enforces network boundaries. Unlisted IP addresses are strictly rejected at authentication time.

4. What is a Permission Set? (Preventing Profile Sprawl)

A **Permission Set** is a collection of settings and permissions used to grant users additional access **without modifying their underlying Profile**.

The Anti-Pattern: Profile Sprawl

Imagine an organization with 100 Sales Representatives sharing the *Sales User Profile*:

- 10 representatives need to Export Reports.

- 5 representatives need to Delete Opportunities.
- 2 representatives need to import Leads via Data Loader (API Access).

If you attempted to solve this strictly through Profiles, you would be forced to create and maintain an absurd permutation of profiles:

```
❌ THE WRONG WAY: PROFILE EXPLOSION (MAINTENANCE NIGHTMARE)
├── Profile 1: Sales User
├── Profile 2: Sales User + Export Reports
├── Profile 3: Sales User + Delete Opportunities
├── Profile 4: Sales User + Export + Delete
├── Profile 5: Sales User + API
└── Profile 6: Sales User + Export + Delete + API  ... (Dozens of clones!)
```

```
✅ THE RIGHT WAY: ATOMIC PERMISSION SETS (MODULAR & REUSABLE)
├── Base Profile: 1 Single "Sales User Profile" (Shared by all 100 users)
├── Perm Set A:   "Export Reports"           (Assigned to 10 users)
├── Perm Set B:   "Opportunity Delete"       (Assigned to 5 users)
└── Perm Set C:   "API Data Import"          (Assigned to 2 users)
```

When a user transfers teams or no longer requires a temporary privilege, the administrator simply unassigns the Permission Set without altering the user's primary profile.

5. The Additive Security Model

One of the most essential rules in Salesforce authorization is that **standard Permission Sets are strictly additive**. They grant access; they *never* revoke or deny permissions already granted elsewhere.

> **Crucial Rule:**
> You **CANNOT** use a standard Permission Set to restrict, revoke, or "take away" an access right granted by a user's Profile!

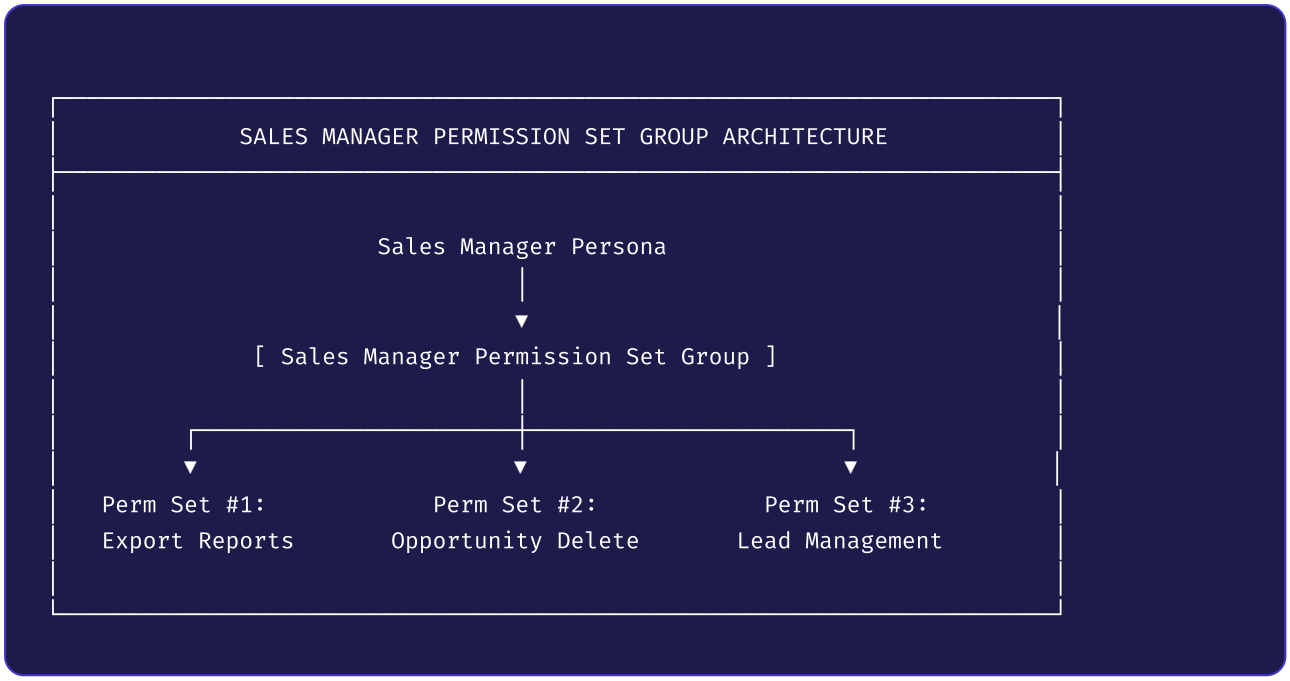
PROFILE SETTING	PERMISSION SET SETTING	EFFECTIVE USER ACCESS	RESULT EXPLANATION
Delete Opportunity = ❌ Disabled	Delete Opportunity = ✅ Enabled	✅ ALLOWED	Permission Set successfully *extends* access.
Delete Opportunity = ✅ Enabled	Delete Opportunity = ❌ Disabled (Unchecked)	✅ ALLOWED	Unchecked in Perm Set does NOT deny access. Profile still permits it!
Delete Opportunity = ❌ Disabled	Delete Opportunity = ❌ Disabled	❌ DENIED	No assigned layer grants the privilege.
Delete Opportunity = ✅ Enabled	Delete Opportunity = ✅ Enabled	✅ ALLOWED	Both layers permit it (redundant but valid).

6. Permission Set Groups (Persona-Based Bundling)

As organizations grow, administering dozens of granular Permission Sets individually becomes tedious. A Sales Manager might need 8 distinct permission sets:

- Export Reports
- Delete Opportunities
- Manage Sales Leads & Lead Conversion
- View All Forecasts
- Mass Update Records

Assigning each permission set manually across dozens of managers invites human error. A **Permission Set Group (PSG)** allows admins to bundle multiple Permission Sets into a single reusable package aligned to a **Job Role / Persona**.



7. Muting Permission Sets (Selective Disabling)

What happens if your *Sales Manager Permission Set Group* includes 5 permission sets (including *Opportunity Delete*), but you have an **Assistant Sales Manager** persona who needs everything in that bundle *except* the ability to delete Opportunities?

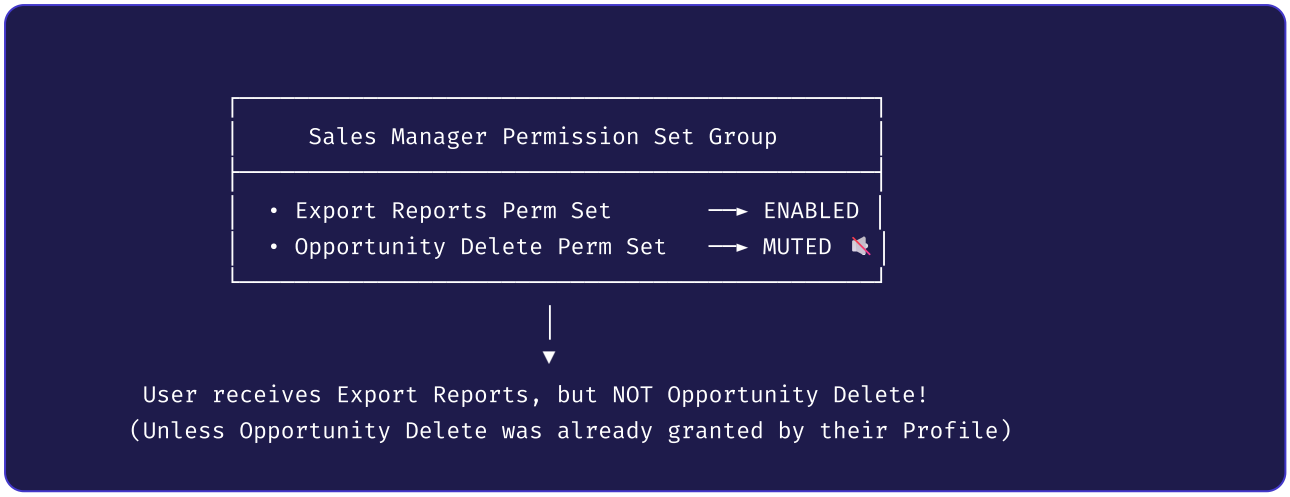
Do you have to dismantle the group or duplicate all 4 other sets into a new group? **No.** Salesforce provides **Muting Permission Sets**.

🔇 How Muting Works

A Muting Permission Set lives *inside a specific Permission Set Group*. When a permission is muted, it zeroes out that specific permission for anyone who receives access via that group.

⚠️ The Boundary of Muting

Muting is **NOT a universal block**! If a user is granted the same permission directly via their Profile or via a standalone direct Permission Set assignment, muting the group *will not* take away that external access.



8. The Grand Comparison Matrix

This matrix summarizes the exact technical boundaries between Profiles, Permission Sets, and Permission Set Groups:

FEATURE / CAPABILITY	PROFILE	PERMISSION SET	PERMISSION SET GROUP
Primary Architectural Purpose	Default base settings & system baseline	Granular, additive task-based permissions	Bundles multiple Permission Sets by persona
Max Assignments per User	**Exactly 1** (Mandatory)	**Multiple** (0, 1, or Many)	**Multiple** (0, 1, or Many)
Object Permissions (CRUD)	Yes	Yes (Additive)	Yes (Combined)
Field-Level Security (FLS)	Yes	Yes (Additive)	Yes (Combined)
System & User Permissions	Yes	Yes	Yes (Combined)
App Settings & Default App	Yes (Sets Default App)	Visibility Only	Visibility Only
Tab Settings	Yes (On / Off / Hidden)	Visibility Extension	Visibility Extension
Record Type Assignments	Yes (Sets Default)	No (Profile-level)	No
Page Layout Assignments	Yes (Exclusive)	No	No
Login Hours & IP Ranges	Yes (Exclusive)	No	No
Can Bundle Sets?	No	No	Yes
Can Mute Permissions?	No	No	Yes (via Muting Set)

9. Hands-On Practical Demonstration (15 Detailed Steps)

Follow these 15 practical steps to implement the complete baseline-to-group security model in your Salesforce Developer Org:

STEP 01

Create Base Profile: "Sales User Profile"

```
Setup ➔ Profiles ➔ [Minimum Access - Salesforce] ➔ Clone
```

In accordance with modern security standards, locate the baseline `Minimum Access - Salesforce` profile (or a custom baseline profile in your developer edition). Click **Clone**, enter Profile Name: `Sales User` (or `Sales User Profile`), and click **Save**.

STEP 02

Configure Baseline Object Access (Opportunities)

```
Setup ➔ Profiles ➔ Sales User Profile ➔ Object Settings ➔ Opportunities ➔ Edit
```

Configure the Opportunity Object Permissions as follows:

- Read: Checked
- Create: Checked
- Edit: Checked
- Delete: Unchecked (Disabled)
- View All / Modify All: Unchecked

Click **Save**. Our baseline sales rep can work with Opportunities but cannot delete records.

STEP 03

Assign "Sales User Profile" to Rahul

```
Setup ➔ Users ➔ Users ➔ Rahul ➔ Edit
```

Change Rahul's Profile dropdown to **Sales User**. Click **Save**. Rahul now inherits our strict baseline access.

STEP 04

Verify Rahul's Baseline Experience

```
Setup ➔ Users ➔ Rahul ➔ Login (as Admin)
```

Navigate to the **Opportunities** tab. Create a test opportunity, update its Stage, and notice that the **Delete** button is completely absent from the page layout and record action menu. Baseline enforced!

STEP 05

Create "Export Reports" Permission Set

```
Setup ➔ Permission Sets ➔ New
```

Enter Label: `Export Reports` (API Name: `Export_Reports`). Leave License as *--None--* (unrestricted) so it can be assigned across compatible Salesforce user licenses. Click **Save**.

Inside the Permission Set, click **System Permissions ➔ Edit**. Check **Export Reports** (Salesforce will automatically check required dependencies). Click **Save**.

STEP 06

Assign "Export Reports" to Priya

```
Permission Sets ➔ Export Reports ➔ Manage Assignments ➔ Add Assignments
```

Select **Priya** from the user list. Click **Assign ➔ Done**. Priya's access now equals: *Sales User Profile + Export Reports Perm Set*.

STEP 07

Demonstrate Permission Revocation

```
Permission Sets ➔ Export Reports ➔ Manage Assignments ➔ Remove Assignments
```

Select Priya and click **Remove Assignments**. Confirm the prompt. Priya immediately loses Export Reports privilege without modifying her Profile or impacting any other team members.

STEP 08

Create "Opportunity Delete" Permission Set

```
Setup ➔ Permission Sets ➔ New
```

Enter Label: `Opportunity Delete`. Click **Save**.
Under *Apps*, select **Object Settings ➔ Opportunities ➔ Edit**.
Under *Object Permissions*, check **Delete** (Read, Create, and Edit will be checked automatically).
Click **Save**.

STEP 09

Negative Baseline Check for Rahul

```
Log in as Rahul ➔ Opportunities ➔ Existing Opportunity Record
```

Confirm once again: Rahul cannot delete opportunities. The action does not appear.

STEP 10

Direct Assignment of "Opportunity Delete" to Rahul

```
Permission Sets ➔ Opportunity Delete ➔ Manage Assignments ➔ Add Rahul
```

Assign the Permission Set directly to Rahul. Switch to Rahul's browser session and refresh the Opportunity record. The **Delete** button immediately appears! Rahul can now successfully delete records.

STEP 11

Create "Sales Manager Permission Set Group"

```
Setup ➔ Permission Set Groups ➔ New Permission Set Group
```

Enter Label: `Sales Manager` (or `Sales Manager Permission Set Group`). Enter Description: *"Persona bundle granting elevated reporting and record management privileges for Sales Managers."* Click **Save**.

STEP 12

Bundle the Two Permission Sets into the Group

```
Sales Manager Group ➔ Permission Sets in Group ➔ Add Permission Set
```

Click **Add Permission Set**. Check both:

- `Export Reports`
- `Opportunity Delete`

Click **Add ➔ Done**. Both sets are now packaged inside our *Sales Manager* group.

STEP 13

Assign the Permission Set Group to Priya

```
Sales Manager Group ➔ Manage Assignments ➔ Add Assignments ➔ Select Priya
```

Assign the entire group to Priya in a single action! She inherits all permissions packaged within both bundled permission sets.

STEP 14

Verify Priya's Combined Access

```
Log in as Priya ➔ Test Reports & Opportunities
```

- Open Reports ➔ Click any Report dropdown ➔ **Export** option is Available.
- Open Opportunities ➔ Open Opportunity record ➔ **Delete** option is Available.

Both capabilities were delivered via **one group assignment**.

STEP 15

Implement a Muting Permission Set in the Group

```
Sales Manager Group ➔ Muting Permission Set in Group ➔ New ➔ Mute Delete
```

Click **New** under Muting Permission Set in Group. Enter Label: `Mute_Opportunity_Delete`. Save.

Open the muting set ➔ **Object Settings** ➔ **Opportunities** ➔ **Edit**. In the **Muted** column, check **Delete**. Click **Save**.

Verification: Log back in as Priya. Since she only received Delete via this group, the Delete privilege is now **muted**! (Meanwhile, if Rahul still has *Opportunity Delete* assigned directly, Rahul can still delete!)

10. The Additive Permissions Truth Table & Troubleshooting Guide

When troubleshooting user access discrepancies during certification exams or on the job, use this comprehensive evaluation logic:

USER PROFILE	DIRECT PERM SET	GROUP PERM SET	MUTING IN GROUP	EFFECTIVE ACCESS
No Access ❌	No Access ❌	Granted ✅	Not Muted ⬜	ALLOWED ✅
No Access ❌	No Access ❌	Granted ✅	**Muted** 🔕	DENIED ❌
No Access ❌	**Granted Directly** ✅	Granted in Group ✅	**Muted in Group** 🔕	ALLOWED ✅
Granted in Profile ✅	No Access ❌	Granted in Group ✅	**Muted in Group** 🔕	ALLOWED ✅
No Access ❌	No Access ❌	No Access ❌	N/A	DENIED ❌

Admin Exam Trap:
Notice row 3 and row 4! If a permission is muted inside a Permission Set Group, but the user is also granted that same permission directly via their Profile or via a standalone Permission Set, the user **STILL HAS ACCESS**. Muting is scoped strictly to the group it resides within!

11. Key Takeaways & Episode Summary

1. One Profile Mandate

Every Salesforce user has exactly one profile. Profiles govern baseline permissions, default apps, record types, page layouts, login hours, and login IP ranges.

2. Additive Permission Sets

Users can have multiple permission sets. Permission sets extend access without multiplying profiles. They cannot restrict or deny access granted by the profile.

3. Persona-Based Groups

Permission Set Groups package multiple permission sets into a single bundle mapped to business personas (e.g., Sales Manager, Regional Director).

4. Scoped Muting Control

Muting permission sets disable specific capabilities within a Permission Set Group without altering the standalone permission sets. Muting does not override external assignments.

12. Next Episode Preview

📺 Coming Up in Episode 35: Object-Level Security & CRUD Deep Dive

In the next episode, we dive deep into **Object-Level Security**! We will thoroughly examine the **CRUD** permissions (**Create, Read, Edit, Delete**) alongside the ultra-powerful administrative overrides: **View All** and **Modify All**. We'll explore how these permissions interact with record ownership and how to bulletproof your data model.
